Exploring Neural Networks for Predicting Next-Pitch Statistics in the MLB

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Introduction

Goal:
This paper looks at how both shallow neural networks and deep neural networks can be used to predict the next pitch in an MLB game, given the game situation.

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Variables

Game Situation Variables:

- Score difference
- Balls
- Strikes
- Outs
- Runner on 1st
- Runner on 2nd
- Runner on 3rd
- Pitch number
- Half inning

Pitch Classes:

- 4-Seam Fastball: FF, FC
- 2-Seam Fastball: SI, FT
- Splitter: FS, FA
- Slider: SL
- Changeup: CH
- Curveball: CU, KC
- Intent. Ball: IN, FO, PO
- Uncommon: EP, AB, KN, SC, UN

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Data

- Kaggle
- 2015 MLB Pitch Data
- Normalized
- Models are 70% Training, 30% Testing, except for FFN (automatic)
- Added: score difference, half innings
- Removed: spin of ball, location of ball, individual team scores, pitcher code
- Baseline accuracy: 12.5%

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Naive Bayesian Classifier

- Assumes conditional independence
- Posterior Probability:

$$P(C_k|x) = P(x_1, x_2, ..., x_D|C_k) \cdot P(C_k)$$

- Assume Gaussion distribution:

$$Norm(x|\mu, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{(0.5)}} exp\left\{\frac{-1}{2\sigma^2}(x-\mu)^2\right\} \quad (1)$$

- Accuracy: 24.25%

Flaws: assuming conditional independence unrealistic in this model, no layers to classify unfamiliar data

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Support Vector Machine (SVM)

- Shallow Neural Network
- MATLAB built in function *fitcauto* automatically finds optimal classification model and hyperparameters
- Accuracy: 40.92%

Flaws: shallow network used for 9 input features, one-versus-one coding risks incorrect scales

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Convolutional Neural Network (CNN)

- Deep Neural Network
- Layers:

```
featureInputLayer(numFeatures,"Name","input")
convolution1dLayer(10,12,"Name","conv1")
fullyConnectedLayer(100,"Name","fc1")
batchNormalizationLayer("Name","bn1")
leakyReluLayer("Name","relu")
fullyConnectedLayer(50,"Name","fc2")
batchNormalizationLayer("Name","bn2")
leakyReluLayer("Name","relu2")
fullyConnectedLayer(numClasses,"Name","fc3")
softmaxLayer("Name","sm")
classificationLayer
```

- Accuracy: 40.52%

Flaws: designed to reduce the number of inputs, a model with only 9 one-dimensional inputs loses a large portion of the data

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Feed Forward Network (FFN)

- Deep Neural Network
- Contains only a specified number of hidden layers between the input and output, 5 for this model
- Activation Function: *sigmoid[0,1]*
- Accuracy: 41.49%

Flaws: one-versus-one coding due to continuous optimization algorithm

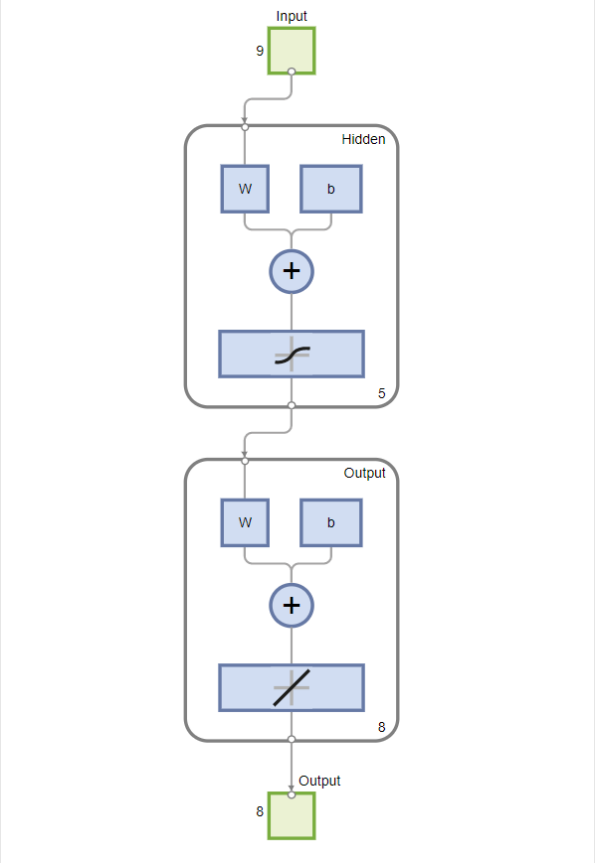


Figure: Feed Forward Network Diagram

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Table: Summary of Model Results

Accuracy Results	
Baseline	12.50%
Naive Bayesian Classifier	24.25%
Support Vector Machine	40.92%
Convolutional Neural Network	40.52%
Feed Forward Network	41.49%

Improved the model:

- Adjusted training rates
- Training, validation, and testing data
- Randomization

Hurt the model:

- Optimization Algorithms
- One-versus-one coding
- Dominating classes

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TABLE III
FFN CONFUSION MATRIX

	1	2	3	4	5	6	7	8
1	288063	0	0	0	0	0	471	0
2	148854	0	0	0	0	0	256	0
3	11214	0	0	0	0	0	13	0
4	101814	0	0	0	0	0	100	0
5	73916	0	0	0	0	0	103	0
6	68797	0	0	0	0	0	41	0
7	2822	0	0	0	0	0	775	0
8	4036	0	0	0	0	0	6	0

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Related Studies

- Glenn Sidle and Hien Tran
  - SVMs, classification trees, linear discriminant analysis
  - Accuracy $= 65.40\%$
- Christopher Watkins
  - Multinomial logistic regression and SVMs
  - Accuracy $= 74.50\%$
- Chih-Chang Yu, Chih-Ching Chang, and Hsu-Yung Cheng
  - Stacked long term and short-term memory model (LSTM)
  - Accuracy $= 76.00\%$

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Future Work

- Break up the model by individual and consistent pitchers - helps with dominating pitch problem
- Weighing the outcomes for attention mechanism
- Increase in features and/or a decrease in pitch type
- Add a mechanism for looking at sequential pitches

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